

Long-term Energy Demand and Carbon-Neutrality Pathways for Egypt

A LEAP-Based Scenario Analysis (2017–2070)

Integrated Energy System Modeling for Sustainable Transition

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Egypt's Energy-Climate Challenge

**116M+**

Population

A rapidly growing population is driving unprecedented increases in energy demand across all sectors.

**6-7%**

Annual GDP Growth

Strong economic expansion fueling industrialization and urban development energy needs.

**92%**

Fossil Fuel Share

Dominant fossil fuel dependency creating exposure to price volatility and high carbon lock-in.

Problem statement: Why This Research Is Needed

! Three Compounding Pressures

01 Surging Energy Demand

Population growth, rapid urbanization, and increasing cooling needs are driving electricity and fuel needs to record levels. **Demand projected to triple by 2050** with no sign of long-term saturation.

02 Fossil Fuel Dependency

Oil and natural gas dominate Egypt's energy mix, creating **exposure to price shocks** and locking in high carbon emissions for decades without transition.

03 Climate and sustainability Commitments

Egypt's **NDC, Vision 2030, ISES 2035, and NCCS 2050** set ambitious renewable and emissions targets, yet integrated pathway implementation remains underexplored.

Research Questions & Research Gap

? Core Research Questions

- Q1** How much can renewable electricity alone reduce Egypt's long-term CO₂ emissions under growing energy demand?
 → Power transition focus
- Q2** How much can end-use efficiency improvements reduce future energy demand and emissions without major power-sector changes?
 → Efficiency transition focus
- Q3** What combined supply- and demand-side pathway enables Egypt to achieve long-term carbon neutrality?
 → Integrated Carbon Neutral Pathway

💡 Research Focus: Future Energy Demand projections to achieve Carbon Neutrality and Improve Energy Efficiency across all sectors.

🔍 What Existing Studies Have Done

- ✓ Partial-sector analyses of Egypt's electricity system (renewable integration in the power sector only)
- ✓ Short-term projections to 2030 or 2035 aligned with Vision 2030 targets
- ✓ Single-scenario studies without a comparative scenario framework
- ✓ Sectoral efficiency or renewable studies without system-wide integration

★ The Gap This Study Fills

No existing study provides a full national, bottom-up, long-horizon (2017–2070) An integrated energy system model for Egypt that jointly evaluates:

Demand-side transformation

Renewable expansion

Electrification

System optimization

→ **Using LEAP soft-linked with NEMO:** developing and comparing six long-term LEAP scenarios to identify realistic carbon-neutrality pathways for Egypt.

Research Goal & Specific Objectives



Overarching Goal

Develop the **first comprehensive, long-range (2017–2070)** integrated energy system model for Egypt to assess carbon-neutrality pathways under **six structured transition scenarios**.

Specific Objectives

O1 Model Development

Build and calibrate a national bottom-up LEAP energy model for Egypt using the **2017 base year** with sector-level techno - economic parameters.

O2 Optimization Integration

Soft-link LEAP with the **NEMO optimization module** to derive a least-cost net-zero benchmark (NZ-OPT) for comparison with policy scenarios.

O3 Scenario Design

Design and run **six transition scenarios** covering the full spectrum from BAU to carbon-neutral pathways through 2070.

O4 Impact Quantification

Quantify the roles of **renewable energy expansion, demand-side efficiency, and electrification** in achieving deep decarbonization.

O5 Policy Insights & Transferability

Provide **evidence-based policy insights** for Egypt and transferable modeling frameworks for other rapidly growing emerging economies in the MENA region and globally.



Research Flow & Methodology

Seven-Step Research Flow

1

Literature Review

LEAP-based studies in MENA & global developing economies
Egypt energy policy: Vision 2030, ISES 2035, NCCS 2050, NDC

2

Data Collection

Base Year 2017: IEA, CAPMAS, EEHC, IRENA, World Bank

3

LEAP Model Setup

Bottom-up demand tree: residential, commercial, industrial, transport, and agriculture sectors

4

Scenario Design

Scenario modelling methodologies & decarbonization pathways
Six transition scenarios from BAU to net-zero pathways

5

NEMO Optimization

Power sector optimization for least-cost net-zero benchmark

6

Analysis & Results

Comparative scenario assessment and impact quantification

7

Policy Implications

Evidence-based recommendations for Egypt and the MENA region

Data Sources & Base Year


IEA

National energy balances, fuel
Consumption by sector, CO₂ emissions


CAPMAS

Population data, household surveys,
economic activity indices


EEHC

Egyptian Electricity Holding Co.
Installed generation capacity, electricity production & consumption


IRENA

Renewable energy technology costs,
capacity factors, deployment potential


World Bank

GDP growth rates, urbanization rates,
transport fleet data


NREA

Solar & wind resource maps, existing
RE projects, pipeline capacity



Model Design & Validation

- LEAP bottom-up demand tree across 5 sectors
- **Supply module: fossil, hydro, solar PV, wind, nuclear**
- **Calibration to 2017 energy balance data**
- **Sensitivity testing for robustness**

Literature Review: Comparison between LEAP Model and other models (bottom-up models)

Park et al. (2016)	Used a bottom-up model to evaluate the electrical energy production in South Korea. Their study involved calculating the optimal portfolio, considering renewable energy technologies explicitly, up until 2050, with the cost effectiveness of measures being assessed.
Prina et al. (2020)	Performed a comprehensive assessment of the categorization algorithms used in modelling bottom-up energy systems. Their findings showed that the Market Allocation and The Integrated Markal-EFOM System (TIMES)(MARKAL/TIMES) and the Open-Source Energy Modelling Systems(OpenSEESYS) were observed to possess somewhat lower temporal and techno-economic data resolutions while exhibiting moderate spatial resolutions.
Prina et al. (2021)	Conducted a thorough analysis of the available bottom-up models, where they evaluated and compared forty-three different studies that utilized these models for analysis at the national level.
Sanchez et al. (2021)	Conducted a study to assess the role of bottom-up energy models in informing policy formulation for enhancing residential building and sector-specific electricity end-use efficiency.
The findings showed that models constructed using a bottom-up approach do not afford a wide range of tools for analysis, which plummets their ability to effectively assist in policy formulation.	

Literature Review : Previous modelling studies globally and for Egypt

Many case studies have been developed on GHG mitigation and reduction policies. Accordingly, there are many researchers who have adopted the LEAP model in their work, and there are several studies that have reviewed Egypt’s energy policies:

Felver, T. B. (2020).	Utilized LEAP to forecast Azerbaijan’s potentialCO2 emissions through three distinct scenarios.
Emodi et al. 2017	Explored Nigeria’s future energy demand, supply, and associated GHG emissions from 2010 to 2040 using the (LEAP) model
Cai et al.2023	Applied LEAP model to investigate different pathways for achieving a carbon emission peak and carbon neutrality by implementing five scenarios before 2030 in China considering economy, transportation, population, power sectors and available renewable sources
Mondal & Ringler .(2020)	Evaluated national and regional benefits of alternative energy development plans, via introducing four different scenarios for the period 2014–2050.these scenarios that involve trading in electricity reduce system costs by 4.5 -7.2% compared to a reference scenario that assumes no energy trading.
Shahid et al.(2021)	Created an LEAP model for Pakistan’s electricity system from 2016 to 2040.
Yang et al.(2021a, b)	LEAP model was developed to examine the ecological and socioeconomic impact of the introduction of renewables in the Chinese province of Zhang.

Energy modelling previous studies for Egypt

Mondal et al. (2018)	Used the TIMES energy system model to examine Egypt’s energy policy goals, as in Egypt’s Vision 2030. concluded that the current supply of energy needs to diversify from natural gas to a mix of technologies and highlighted opportunities to fulfill rising electricity demand while addressing energy security and low-carbon development concerns from 2016 to 2050
Rady et al.2018	Used OSeMOSYS to model Egypt’s power-generation sector, to provide policymakers with better plan for future investments in the power sector. It concluded that the lowest-cost electricity-generation mix always includes hydropower, natural gas-fired steam cycles, wind power, and solar PV rooftop technologies, focused on the analysis of two different energy demand scenarios and did not investigate increasing RETs for power generation in Egypt.
Elsayed et al.(2023)	Analyzed various scenarios of electrical energy generation and consumption in Egypt for 2020 and 2050 using LEAP model . The results show that the natural gas contribution will decline till reaching its peak of 61.5% by 2050 for (NON) scenario. However, this share will be less than 16.5% by 2045 for (REN.70) scenario. Whilst nuclear share will be 4.6% by 2050 for (REF.A) scenario. Found that the renewable scenarios have the lowest long-term production cost compared to other scenarios. However, this study does not consider deep detailed analysis for energy demand.
Gibson et al.(2024)	Included a renewable energy development scenario based on ISES 2035 targets and IRENA’s proposed modifications.
Mondal et al. (2019)	Examined Egypt’s power sector energy supply strategies using the TIMES model, with a particular emphasis on CO2 mitigation, renewables integration and restrictions on natural gas usage.

Findings : indicated that energy mix diversification is necessary to improve supply reliability, reduce reliance on fossil fuels and reducing GHG emissions.

LEAP Model — Framework Overview



What is LEAP?

What Is **LEAP**?
(A quick reminder)

What does LEAP do?

LEAP (Long-range Energy Alternatives Planning) is a widely-used bottom-up integrated energy planning tool developed by the **Stockholm Environment Institute (SEI)**.

- ✓ Models' energy demand, supply, transformation, and GHG emissions
- ✓ Bottom-up approach models each sector & end-use technology
- ✓ Scenario flexibility: any number of user-defined pathways
- ✓ Full GHG accounting across all fuels and sectors

NEMO Soft-Link

NEMO (Next Energy Modelling System for Optimisation) is an open-source, least-cost power sector optimization model.

Soft-linked with LEAP, NEMO determines the **optimal technology mix, capacity expansion, and dispatch schedule** for Egypt's electricity system under each scenario's constraints.

Egypt LEAP Model Structure

DEMAND SECTORS



Residential



Commercial



Industrial



Transport



Agriculture



TRANSFORMATION



Power Plants



Oil Refineries



Gas Processing



Transmission



RESOURCES / SUPPLY



Natural Gas



Oil



Solar PV



Wind



Hydro



Nuclear

Base Year: 2017 | Time Horizon: 2017–2070 | Soft-linked with: NEMO | 6 Scenarios

06 | LEAP Model — Equations

As a result of these features, the LEAP model is best suited for creating scenarios for complex energy systems. (Malka et al.2021)

- ✓ The energy consumption for each activity level broken down by subsectors and fuel types is calculated in the model using Eq. 1

$$ED_k = \sum_i \sum_j AL_{k,j,i} \bullet EI_{k,j,i}$$

ED_k : stands for energy consumption,
 AL : stands for activity level,
 EI : energy intensity.

EI : Energy used by fuel type k , technology j , in sector i per unit of activity for sector / technology

$$EI_{k,j,i} = \frac{\text{Final energy consumption of (fuel } k, \text{ tech } j, \text{ sector } i) \text{ in base year}}{\text{Activity level of sector or technology in base year}} = \frac{ED_{k,j,i}}{AL_{k,j,i}}$$

- ✓ The carbon emission for every activity was estimated using Eq. (2):

$$ED_{E,L} = \sum_i \sum_j \sum_k AL_{k,j,i} \bullet EI_{k,j,i} \bullet EF_{k,j,i,P}$$

- $ED_{e,i}$ emission is the pollutant emission,
- AL refers to the activity level,
- EI refers to the energy intensity, and
- $EF_{k,j,i,p}$ is the pollutant P emission (kg) factor from fuel type k for equipment j , from sector i .

- ✓ The emission factor per each fuel type is calculated by using Eq. (3)

$$EF = NCV \bullet CEC_u \bullet COF \bullet 44/12$$

EF stands for the localized carbon emission factors,
 NCV stands for net calorific value, CEC_u stands for the carbon oxidation rate of energy u ,
and $44/12$ stands for the molecular weight ratio of CO_2 and C .

LEAP Model — Mathematical Framework

“ As a result of these features, the LEAP model is best suited for creating scenarios for complex energy systems. (Malka et al. 2021)

1 Energy Consumption

$$ED = AL \times EI$$

Energy Demand = Activity Level × Energy Intensity

ED_k: Energy consumption by fuel type k

AL: Activity level (demand driver)

EI: Energy intensity (energy per unit activity)

Calculates energy consumption for each activity level broken down by subsectors and fuel types

2 Carbon Emissions

$$\text{Emission} = AL \times EI \times EF$$

Emissions = Activity × Intensity × Emission Factor

ED_{e,i}: Pollutant emission from sector i

AL: Activity level

EI: Energy intensity

EF_{k,j,i,p}: Emission factor for pollutant p from fuel k, equipment j, sector i

Estimates carbon emissions for every activity across all sectors and fuel types

3 Emission Factor

$$EF = NCV \times CEC \times (44/12)$$

Localized carbon emission factor calculation

EF: Localized carbon emission factors

NCV: Net calorific value of fuel

CEC_u: Carbon oxidation rate of energy u

44/12: Molecular weight ratio of CO₂ to C

Calculates emission factor per fuel type, incorporating local energy characteristics

Six Transition Scenarios

From Business-as-Usual to Least-Cost Net-Zero Pathway

S1

BAU / REF

Business-as-Usual Reference

Baseline projection with **no additional policies**. Historical trends in energy demand, fuel mix, and emissions continue to 2070.

🕒 Serves as reference for all scenario comparisons

S2

PET

Policy-Aligned Transition

Reflects Egypt Vision 2030, ISES 2035, NCCS 2050, and updated NDC commitments. Models officially planned RE targets.

🏛️ Policy-consistent transition pathway

S3

RPT

Renewable Power Transition

Aggressive large-scale integration of solar PV, wind, and hydro in the power sector. Deep decarbonization of electricity supply.

⚡ Supply-side decarbonization focus

S4

EDT

Efficiency-Driven Transition

Emphasis on demand-side energy efficiency improvements and demand moderation across residential, commercial, industrial, and transport sectors.

💡 Demand-side mitigation focus

S5

CNS

Carbon-Neutrality Strategy

Integrates high renewable penetration with demand-side measures under a long-term carbon emissions constraint. Coordinated deep decarbonization.

★ Integrated net-zero pathway

S6

NZ-OPT

Least-Cost Net-Zero (NEMO)

LEAP is soft-linked with NEMO to derive the least-cost pathway to net-zero. Optimal technology mix, investment sequencing, and system flexibility.

📊 Cost-optimization benchmark



Scenario Logic: Progressive isolation of policy effects, supply-side decarbonization, demand-side efficiency, and their combined role in achieving carbon neutrality

09 Egypt: National LEAP Scenario Framework (2017–2070)			
Scenario	Aim	Description	What Changes vs. BAU
Reference / Current Policies (BAU / REF)	baseline benchmark under continuation of existing and already-announced energy and climate policies. Provide a baseline benchmark under continuation of existing and already-announced energy and climate policies.	Gradual renewables growth but continued dominance of fossil fuels, especially natural gas, in power and final energy use. Rising energy demand, only incremental efficiency gains, and steadily increasing energy-related CO ₂ emissions to 2070.	Demand grows with population & GDP; fossil fuels (esp. gas) remain dominant; no additional decarbonization effort Fossil fuel lock-in, high and rising CO ₂ emissions, exposure to fuel price volatility and import risks, and misalignment with long-term climate and sustainability goals.
Planned Energy Transition PET	Full implementation of Egypt Vision 2030-35, NDCs, and official energy strategies : Assess the impact of fully implementing current national energy and climate strategies and NDCs (e.g. Vision 2030).	Policy-consistent transition with faster renewables deployment,	Faster renewables, moderate electrification, stronger efficiency than BAU; Fossil fuels, especially gas, remain important. emissions growth slows but does not reverse Implementation and financing gaps, institutional and regulatory barriers, risk that announced targets and plans are delayed or only partially delivered.
Renewable - Led Power Transition RPT	Supply-side decarbonization case: Explore an accelerated decarbonization pathway focused on the power sector.	Accelerated solar & wind deployment in the power sector :High solar and wind penetration dominate the future electricity mix (optionally under a power-sector CO ₂ cap). Supported by grid reinforcement, storage and flexibility options. Limited new fossil capacity and earlier fossil retirements. Power-sector CO ₂ falls sharply; electricity becomes a key low-carbon carrier for end-use sectors.	High penetration for Wind & solar Rapid fossil phase-down in electricity; grid expansion, storage, electrification of end-uses: High upfront capital needs, grid integration and system flexibility challenges, resistance from fossil incumbents, and need for strong planning, regulation, and system operation capabilities.
Efficiency-Driven Demand Transition EDT	Demand-side decarbonization case : Assess the potential of ambitious cross-sectoral energy efficiency to curb demand and emissions.	Aggressive & strong efficiency improvements across buildings, industry, and transport (reduction in energy demand growth : Strong end-use efficiency in buildings, industry and transport: higher appliance and building standards, industrial process upgrades, efficient vehicles and public transport, plus some electrification. Final energy demand growth is substantially reduced relative to BAU, with only moderate supply-side transformation.	Lower energy intensity; slower final energy demand growth; limited supply-side change : High initial investment needs, behavioral and institutional barriers, split incentives (e.g. landlord–tenant), difficulties enforcing standards and achieving widespread behavioral change.
Carbon-Neutrality Strategy (Narrative) CNS	Main net-zero pathway: Define a policy-driven, narrative pathway consistent with achieving economy-wide net-zero CO ₂ by 2070.	Combined supply- and demand-side transformation: deep efficiency, extensive electrification, near-zero-carbon power, fuel switching, and phase down of unabated fossil fuels after mid-century. Limited role for CCS and low-carbon fuels in hard-to-abate sectors. Energy demand is much lower than BAU and CO ₂ declines steeply toward near-zero by 2070.	Deep efficiency, high electrification, near-zero-carbon power, fossil phase-out after mid-century: Very high investment and coordination requirements, technology and policy uncertainty, social and labour transition challenges, dependence on long-term political commitment and international support.
Net-Zero Optimized Pathway (NEMO) NZ-OPT	Provide a cost-minimizing benchmark Net-Zero pathway under an economy-wide CO ₂ constraint to 2070.	NEMO optimizes the energy system under a long-term CO ₂ cap that drives energy-related emissions to net-zero by 2070. Chooses the least-cost mix of generation, conversion, and end-use technologies subject to technical and policy constraints. Serves as an optimization benchmark to compare with the narrative CNS pathway.	Optimization-driven technology mix; serves as comparison to CNS (not a policy narrative) Strongly dependent on assumptions (technology costs, resource potentials, constraints). May yield pathways that are cost-optimal in the model but politically or socially difficult to implement or misaligned with just transition priorities

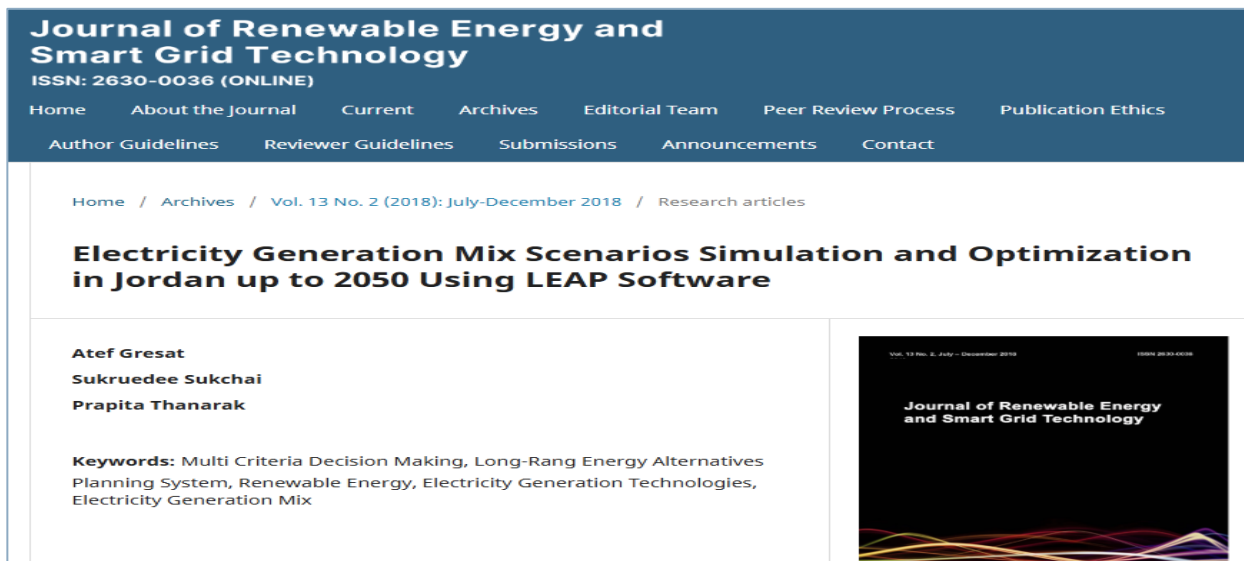
Scenario	Main Purpose	Electricity Mix Assumptions	Demand & Energy Assumptions	Emissions Logic
S1 BAU / REF Reference	Baseline benchmark under continuation of existing policies	Gas-dominated system; RE 30–35% by 2050	Final energy demand growth: 2.0–2.5%/yr; Efficiency improvement: 0.5%/yr	CO ₂ emissions rise until ~2040, then stabilize
S2 PET Policy- Aligned	Vision 2030, Energy Strategy 2035, NDC implementation	RE share 42% by 2035; 45–50% by 2040	Demand reduction vs BAU: 5 – 8% by 2030; Efficiency improvement: 1%/yr	CO ₂ emissions peak 2030–2035, gradual decline after
S3 RPT Power Transition	Clean electricity focus; isolate power sector decarbonization	RE share 60% by 2040; 75 –80% by 2050; 85 – 90% by 2070 Gas plants remain as backup/reserve capacity with low utilization for system adequacy and flexibility.	Final energy demand follows BAU (no extra efficiency)	Power-sector CO ₂ ↓ 70 – 85% by 2050; residual emissions remain
S4 EDT Efficiency- First	Demand-side mitigation; isolate efficiency impact	Power sector follows PET scenario; limited supply-side change: so decarbonization is driven primarily by demand-side measures rather than supply-side change.	• Buildings' energy intensity ↓ 30 – 40% by 2050; Transport energy/km ↓ 25–35%; Appliances: Best Available Technology by 2040. Final energy demand vs BAU ↓ 15 - 20 % by 2040 to ↓ 20 – 25 % by 2050	CO ₂ emissions reduced, but fossil power remains
S5 CNS Carbon - Neutrality	Integrated net-zero pathway; combined effect under CO ₂ constraint	RE share 80 – 90% by 2050; 90 – 95% by 2070	Final energy demand ↓ 25 – 30% vs BAU by 2050; Strong electrification in buildings & transport	Net emissions → near zero by 2070
S6 NZ- OPT NEMO Optimized	Least-cost net-zero benchmark; cost-optimal reference : NZ-OPT is a cost-optimal benchmark that shows the cheapest way to reach net-zero, used to test whether our policy-driven pathway is realistic and efficient.	Model chooses optimal mix under constraint	Model selects cost-optimal efficiency & supply	Hard net-zero constraint by 2070; emissions must be zero

💡 **Key Insight:** "CNS represents a realistic policy pathway, while NZ-OPT provides an objective benchmark to assess its economic efficiency."

Scenario Comparison → Egypt Framework

Country	Scenario Focus	Key Strength	Main Limitation	Lesson Used for Egypt
Japan	Economy-wide net-zero under CO ₂ constraints	Clear net-zero ambition; constraint-based logic	Heavy reliance on imports, CCS/CDR, high costs	Net-zero framing & least-cost benchmark (CNS, NZ-OPT)
Jordan	Power-sector transition (LEAP)	Clear renewable-led electricity pathways	Demand side largely implicit	Control power-sector scenario/ NOT rely on demand reductions (RPT) :
Korea	System-wide supply –demand transition	Strong efficiency & electrification treatment	Highly industrialized, low growth context	Depending on demand-side pathway (efficiency of end uses) / power sector close to BAU → (EDT)
Spain	Policy-driven decarbonization (LEAP)	Clear baseline → policy → decarbonization structure	Assume EU regulations, stable energy demand, which differs from Egypt’s growing demand	Policy-aligned transition (PET) (Full implementation of Egypt’s existing energy and climate policies)

Jordan Scenarios: LEAP-Based Electricity Scenario Framework



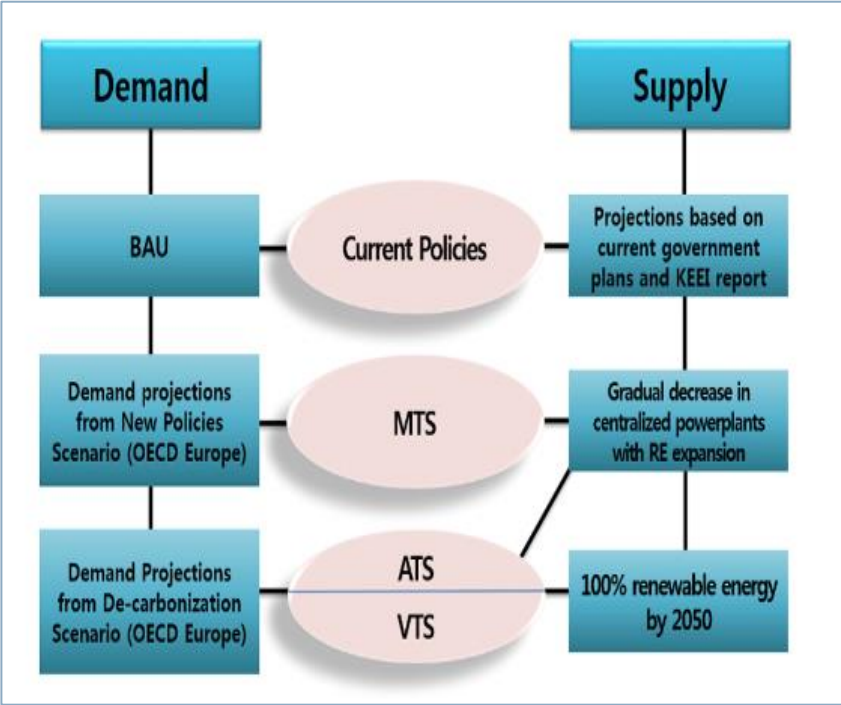
- It provides a clear, LEAP-based framework focused on the electricity sector.
- Jordan scenarios follow a logical progression from a business-as-usual baseline to renewable expansion and finally to a CO₂ - constrained case.
- This structure is for understanding power-sector decarbonization pathways and the role of renewables in reducing emissions.
- The main strength of the Jordan framework is its clarity in analyzing clean power-sector transitions.
- **Key limitation:** demand-side dynamics and energy efficiency across end-use sectors are not explicitly addressed.

Gresat, A., Sukchai, S., & Thanarak, P. (2018). *Journal of Renewable Energy and Smart Grid Technology*, 13(2).

scenario	Focus	Description
Scenario 1	BAU (base line)	Continuation of current electricity expansion plans Business-as-usual (base plan to 2025; conventional replacement thereafter)
Scenario 2	BAU + RE (Policy transition)	Renewables added to meet future demand growth: BAU to 2025 + renewables added after 2025 to meet demand (no new conventional replacements)
Scenario 3	High – RE (BAU + Nuclear) Supply-side decarbonization	Renewables dominate electricity generation BAU to 2025 + two nuclear plants + new gas plants to meet demand
Scenario 4/5	(coal / oil shale variants)	Adds coal and/or oil shale technologies
Scenario 6	CO₂-constrained CO₂ limit	Climate constraint : Electricity expansion limited by emission caps Same as prior scenario but with CO₂ emissions restrictions/constraints

South Korea: LEAP-Based Whole Energy System Scenarios

Code	Scenario	Focus	Description
BAU	BAU/Baseline	Existing policies and strategies	Energy demand and supply follow existing policies, limited renewable expansion
MTS	Moderate Transition Scenario	Policy-driven transition with increased RE shares and moderate demand management	Moderate renewable growth and efficiency improvements
ATS	Advanced Transition Scenario	Supply-side decarbonization + electrification	Accelerated renewable expansion with deeper structural change in the power sector and electrification
VTS	Visionary Transition Scenario	High-ambition transition combining clean supply and reduced energy demand	Clean supply combined with reduced energy demand no reliance on crude oil, coal, natural



Summary of the Korea scenarios

- Korean LEAP 4 scenarios provide a structured example of energy transitions based on energy consumption and RE share differences up to **2050**.
 - Scenarios allows to interaction between supply-side decarbonization and demand-side transformation **to be assessed within a** single modelling framework.
 - It emphasizes changes in both **renewable energy share and total energy demand**, which aligns well with Egypt’s proposed scenario structure.
 - Adapt to Egypt by distinguishing policy-driven, renewable-led, and efficiency-driven pathways within a consistent modelling framework.
- reflects a highly industrialized economy, which means that some assumptions are not directly transferable to Egypt.

South Korea scenarios using LEAP model



Energy Policy

Volume 127, April 2019, Pages 425-437



Long-term energy strategy scenarios for South Korea: Transition to a sustainable energy system

Jong Ho Hong^a , Jintae Kim^a, Wonik Son^a, Heeyoung Shin^a, Nahyun Kim^a, Woong Ki Lee^b,
Jintae Kim^c

Highlights

- This paper proposes three scenarios for South Korea's sustainable energy transition.
- This study developed transition scenarios toward renewable energy for both supply and demand.
- Transition pathways were modelled using LEAP up to the year 2050.
- The alternative scenarios show a more successful transition than the BAU scenario.

Hong et al. (2019), *Energy Policy* : this article develops multiple energy system scenarios using the **LEAP model** for South Korea.

Hong J. H., Kim J., Son W., Shin H., Kim N., Lee W. K., & Kim J. (2019). *Long-term energy strategy scenarios for South Korea: Transition to a sustainable energy system. Energy Policy*, 127, 425–437

Japan's Net-zero emissions pathways /scenarios



Energy and Climate Change

Volume 5, December 2024, 100128



Mid-century net-zero emissions pathways for Japan: Potential roles of global mitigation scenarios in informing national decarbonization strategies

Ken Oshiro^a  , Shinichiro Fujimori^{a b c}

Highlights

- Electrification and carbon dioxide removal are essential decarbonization options.
- Demand reduction and hydrogen use are effective to reduce carbon dioxide removal.
- Import of hydrogen-based energies is an option but leads to import dependency.
- International emission trading is an option but could worsen the trade deficit.
- Global models can inform national scenarios associated with international trade.

Oshiro, K., & Fujimori, S. (2024). *Mid-century net-zero emissions pathways for Japan: Potential roles of global mitigation scenarios in informing national decarbonization strategies*. *Energy and Climate Change*.

Ken Oshiro & Shinichiro Fujimori (2024), *Energy and Climate Change* ; develop long-term net-zero decarbonization pathways for Japan and evaluate how global climate scenarios can inform national strategies.

Japan's approach combines **global climate ambition** with **national policy /strategy instruments**

- Explores diverse pathways for Japan to achieve its net-zero emissions target by 2050, considering both domestic and international mitigation options, a national bottom-up energy system model (AIM/Technology-Japan) with global IAM outputs.

Five NZE pathways:

- CDR-heavy (BECCS + DACCS)
- Alternative fuels (hydrogen, synfuels)
- Low-demand society
- Import-dependent (hydrogen & synfuels)
- Emission trading (international credits)

Key findings include:

- **Electrification + power sector decarbonization** are unavoidable across all scenarios.
- Carbon Dioxide Removal (CDR), and hydrogen/synthetic fuels are essential decarbonization options across all net-zero scenarios.
- Upscaling CDR, reducing energy demand, and importing low-carbon energy carriers or emission allowances are alternative pathways, each with their own technological, economic and societal challenges.
- Global energy system models can inform national scenario assessments by providing information on international trade of low-carbon energy and emission allowances, which are important options for Japan given its limited domestic energy resources(domestic renewables are insufficient).

Morocco's OSeMOSYS-based scenario modelling transition to renewable power



Energy Conversion and Management
Volume 317, 1 October 2024, 118857



Research Paper

Towards a sustainable energy future: Modeling Morocco's transition to renewable power with enhanced OSeMOSYS model

Jabrane Slimani ^a, , , Abdeslam Kadrani ^a, Imad El Harraki ^b, Elhadj Ezzahid ^c

Highlights

- Morocco could achieve a 92% RE integration rate by 2050 with an additional \$32 billion total cost.
- Implementing **EE measures** can reduce energy demand by 15% from 2030 to 2050 compared to baseline forecasts.
- The NANES scenario results in the lowest emissions rate of 0.29 MtCO₂e by 2050.
- A comprehensive policy framework tailored for developing countries is proposed.
- Advanced planning tools and an adapted OSeMOSYS model improve energy resource management in volatile conditions.

Goal : Develops an enhanced OSeMOSYS energy system model to examine long-term energy supply strategies (reach a **renewable - dominant electricity system** by 2050

- A baseline and three alternative scenarios were developed,

How far can decarbonize electricity if the renewables increase and slightly control demand?

Scenario	Objective	Key Assumptions
BAU	Continuation of current policies	• Electricity demand keeps rising • No new CO₂ constraint; • Delayed gas investments (still needed) • Moderate RE growth
RNES (Ratified National Energy Strategy)	Policy-aligned transition(official policy)	• Renewables reach 52% of capacity by 2030 • Small demand reduction (5%) vs baseline
NMNES (New Moderate National Energy Strategy) (Moderate Decarbonization)	Accelerated RE + emissions control	• Renewables > 60% by 2050 (57% 2040,67% 2050) • Demand grows slower (reduction 10% lower than BAU) • CO₂ reduction from 2025;
NANES (New Ambitious National Energy Strategy) (Ambitious Decarbonization)	Near-zero electricity emissions	• 72 % RE by 2040, 92 % by 2050; • Strong demand reduction ~15 % • Strong CO₂ constraint • Storage & flexibility required

Spain's energy scenarios and driving forces of CO₂ emissions



Energy Reports

Volume 13, June 2025, Pages 4563-4585



Research paper

Exploring energy scenarios and driving forces of CO₂ emissions of regions: The case of a southern region of Spain through LEAP analysis

Lucía Galán-Cano ^a, Juan Cámara-Aceituno ^a, Julio Terrados-Cepeda ^a,
Manuel Jesús Hermoso-Orzáez ^a , Edgar Antonio Barragán-Escandón ^b,
Ángel Mena-Nieto ^{c d}

Highlights

- LEAP applied to energy planning in southern Europe, using a region in southern Spain.
- Energy and CO₂ scenarios for 2030/2050 in southern Europe analysed with LEAP.
- Highlights renewables' potential to reduce energy demand and CO₂ emissions regionally.
- Electrification and storage identified as key pillars for regional climate neutrality.
- Provides a transferable framework for energy planning in Mediterranean regions.

Key points :

- Compares 3 scenarios : Baseline (current policy); Efficiency scenario (ambitious energy saving actions) ; Long-term decarbonization (targeting climate neutrality) pathways for energy demand and CO₂ emissions
- Finds that even ambitious efficiency improvements and renewable integration may not fully achieve long-term climate targets without deeper transformation (e.g., electrification, storage).

Kenya's Low Carbon Futures by 2050



Article

Kenya's Low Carbon Futures: An Assessment Using the KCERT Model

Jeremy Gachanja ¹, Betsy Muriithi ¹, Onesmus Mwabonje ^{2,*}, Alvin Mugwe ¹, John Olukuru ¹, Izael Pereira Da Silva ³, Francis Mwangi ⁴, Alexandre Strapasson ⁵ and Paisan Sukpanit ²



It aims to **explore long-term low-carbon development pathways** for Kenya up to 2050 by Estimating future energy demand and emissions

- Testing policy and mitigation options
- Supporting national climate strategy and NDC planning

- **KCERT (Kenya Carbon Emissions Reduction Tool)** → Built on **LEAP** framework
- **Scenario-based simulation model** covering the **whole energy system**

Scenarios

- **BAU (Business-as-Usual)** → Continuation of current trends
- **Low-Carbon Development Scenario** → Strong renewables, efficiency, electrification
- **High Ambition Mitigation** → Aggressive mitigation across all sectors
- **Sector-Focused Mitigation** → Targeted actions in power, transport, buildings
- **Sensitivity / Policy Variants** → Variations in technology costs, growth, and uptake rates

Findings :

1. Kenya can achieve **large emissions reductions** by : Expanding geothermal and renewables , Electrifying transport and Improving efficiency
2. Many mitigation options are: **Cost-effective** or **low-cost**
3. Power sector becomes: reach to low-carbon early due to geothermal and hydro

Limitations

1. **No explicit net-zero constraint** so No “hard carbon-neutrality by year X” scenario
2. **No structural separation of pathways** : Power-only vs demand-only vs combined strategies
3. **Not an optimization benchmark** , cannot provide a least-cost net-zero reference like your **NZ-OPT (NEMO)**

Ethiopia's Scenarios using OSeMOSYS- LEAP model



Energy Strategy Reviews

Volume 45, January 2023, 101045



Long-term electricity supply modelling in the context of developing countries: The OSeMOSYS-LEAP soft-linking approach for Ethiopia

Dawit Habtu Gebremeskel ^a , Erik O. Ahlgren ^b, Getachew Bekele Beyene ^a

To analyze long-term **electricity supply** pathways for Ethiopia Least-cost **capacity expansion** under different development strategies, using OSeMOSYS with LEAP which used mainly for demand projection ,focus on:

- Power system expansion
- Hydropower and renewable utilization
- Investment needs and system costs
- Regional electricity exports

Findings : Ethiopia can supply very low-carbon electricity mainly using:

- Hydropower
- Geothermal Wind
- Export-oriented strategies require : Massive grid and generation investment
- Diversification is important to: Reduce drought risk & Improve system reliability

Highlights

- A soft-linking approach of coupling two modelling frameworks (LEAP and OSeMOSYS) is adopted in this study.
- Context and unique features of developing countries are considered during modelling.
- Grid-extension and off-grid supply options are assessed in improving the low-electricity access in Ethiopia.
- Renewable technologies are more competitive and favourable in the context of Ethiopia.
- The improved efficiency scenario has resulted in 11% total discounted cost saving.

Scenarios

- 1.Reference / BAU → Current plans and trends continue
- 2.High Demand / High Growth → Rapid domestic electricity demand growth
- 3.Export-Oriented → Large-scale generation built to export power to neighboring countries
- 4.Hydropower-Dominated → Maximum exploitation of hydro potential
- 5.Diversified Renewables → Mix of hydro, wind, solar, geothermal to reduce climate and hydrology risk

Limitations

1. Only electricity sector : No buildings, transport, industry, or demand-side efficiency
2. No economy-wide emissions : No carbon-neutrality or net-zero constraint
3. No demand-side mitigation: Demand is exogenous; no efficiency scenarios

South Africa's pathway to net zero emissions by 2050



Energy for Sustainable Development
Volume 79, April 2024, 101401



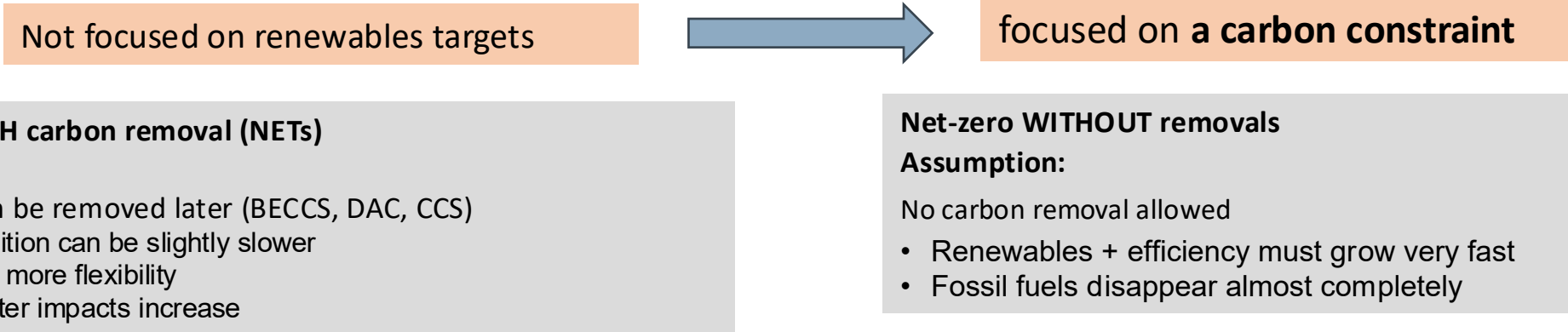
Role of negative emission technologies in South Africa's pathway to net zero emissions by 2050

Sandylove Afrane ^{a, b}, Jeffrey Dankwa Ampah ^c  , Abdulfatah Abdu Yusuf ^d, Zhao Jinjuan ^a, Pingjian Yang ^b  , Jian Lin Chen ^e, Guozhu Mao ^a

Highlights

- Study employs GCAM to model South Africa's net-zero ambitions.
- The role of NETs on the country's energy-land-water resources is quantified.
- Coal power generation declines by 85–99% by 2050 under such ambitions.
- Scenarios with BECCS could decrease cropland and pasture by 59–69% by 2050.
- 0.02–0.1 km³ of water demand by 2050 for energy generation and negative emissions.

- ✓ Uses the the Global Change Assessment Model (GCAM-South Africa) integrated assessment model to explore long-term decarbonisation pathways up to 2050.
- ✓ Considers both emissions reduction measures and NET deployment (e.g., BECCS and DACCS) in its scenario set.
- ✓ Formulates **five distinct scenarios**:
 1. A **reference** scenario that does not incorporate climate policy
 2. **Four net-zero scenarios** with varying assumptions about NETs and mitigation strategies.



Country	Model & Focus	Scenario Type	Strengths	Limitations	What Egypt Builds On
Kenya	KCERT / LEAP (economy-wide, policy tool)	BAU vs Low-Carbon policy narratives	Good sector coverage; strong policy storytelling	No separation of supply vs demand effects; no hard net-zero constraint; horizon to 2050	Scenario communication, policy linkage, sectoral presentation
Ethiopia	OSeMOSYS + LEAP (electricity planning)	Power-sector development & export strategies	Strong capacity-expansion and cost optimization	Only electricity sector; no demand-side mitigation; no economy-wide decarbonisation	Power-system modelling, resource constraints, investment logic
Morocco	OSeMOSYS (power system, decarbonization)	BAU → policy → moderate → ambitious RE	Clear RE targets, demand wedges, CO ₂ pressure, flexibility needs	Electricity-focused; not economy-wide; no full demand-side transition	Scenario architecture: RE targets + demand wedges + CO ₂ pressure
Egypt	LEAP + NEMO (full energy system)	BAU / PET / RPT / EDT / CNS / NZ-OPT	Separates policy, supply, and demand effects; economy-wide; long-term (2070); net-zero + cost-optimal benchmark	?????	Integrates and extends African best practice into a diagnostic, policy-relevant framework

Explicit Mapping: Africa → Egypt Scenario Framework

Egypt Scenario

BAU / REF

What if Egypt continues as today

PET (Policy-aligned transition)

What if Egypt actually does Vision 2030 & NDC?

RPT (Power transition)

What if Egypt clean electricity as fast as possible?

EDT (Efficiency-first)

What is happen If buildings and transport, other appliances use less energy

CNS (Carbon -neutrality strategy)

What if Egypt combines everything needed?

NZ-OPT (NEMO)

What is the cheapest way to reach net-zero?

Other scenarios

Morocco – BAU

Morocco – RNES

Morocco – NMNES / NANES

Morocco – Demand reduction wedges

Morocco – NANES + South Africa net-zero logic

South Africa – Net-zero with/without NETs

Key assumptions for Egypt

Continuation of current policies (baseline under growing demand)

- Renewables grow slowly
- no additional CO₂ constraint
- Gas remains dominant

Full implementation of official national targets and plans (Egypt Vision 2030 and NDC commitments)

- Official renewable targets
- Official efficiency improvements
- No extreme assumptions

Accelerated renewable deployment with milestone targets (Egypt’s strong solar and wind potential)

- High solar & wind growth
- Gas declines gradually
- Demand stays as BAU no change

Explicit demand reduction relative to baseline (5 – 15%) (efficiency gains in buildings, appliances, and transport)

- Efficient appliances
- Better buildings
- Efficient transport
- Power mix not radically changed

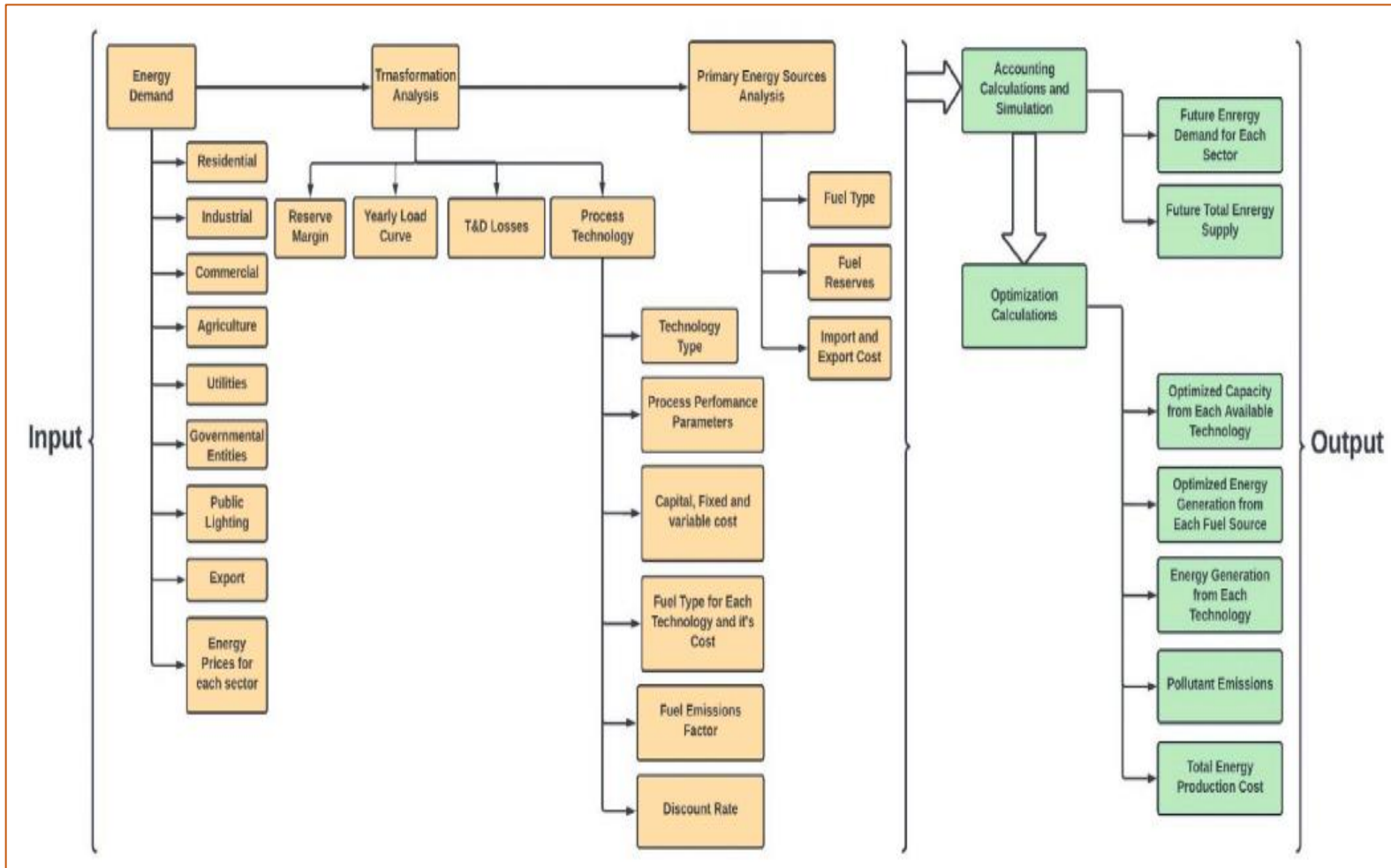
Combined RE expansion, demand moderation, and CO₂ constraint (a realistic, narrative pathway to carbon neutrality)

- Clean electricity
- Strong efficiency
- Long-term CO₂ constraint

Hard net-zero constraint shaping system optimisation (Provides a least-cost benchmark under strict emissions limits)

- Emissions must reach zero (or near-zero)
- Model chooses technologies
- No narrative bias

LEAP framework for Egypt



LEAP model for Egypt

Scenario development

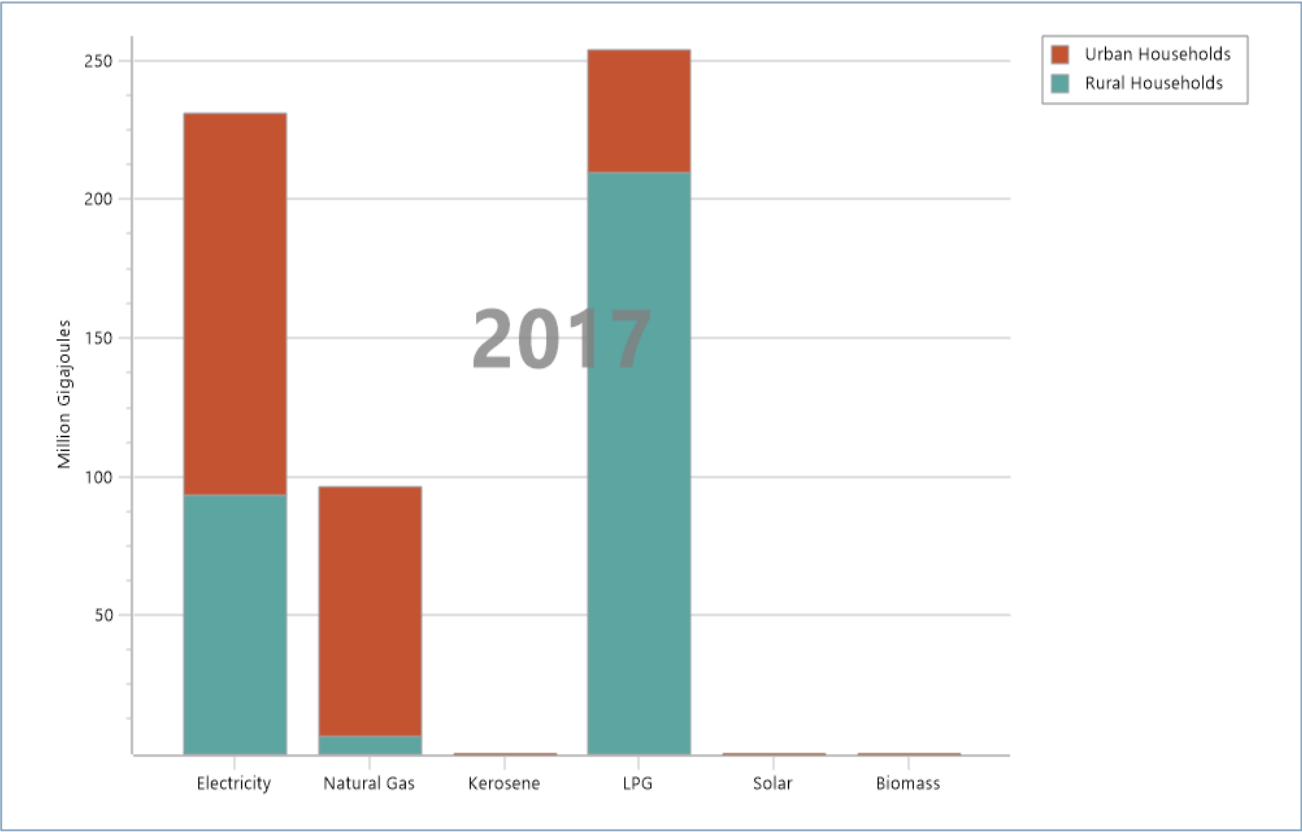
- ✓ Examining energy demand, which informs resource energy and transformation analyses.
- ✓ Assessing available energy supply resources and technologies for converting energy sources, including fossil fuels and renewables
- ✓ Calculating projected energy consumption based on demand, followed by optimization that employs cost-benefit analysis.

Scenario design and analysis

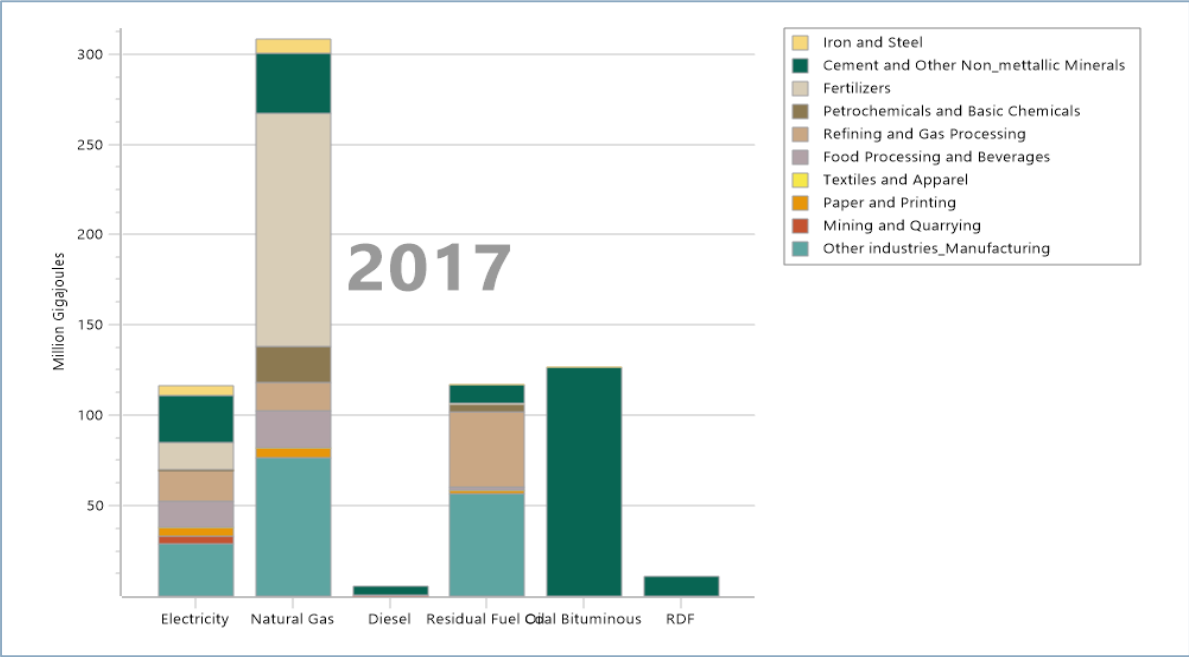
1. The year 2017 is the baseline year, and 2070 is the end year of analysis.
2. Three scenarios will be developed in the LEAP Model
 - Current account, its base year 2017
 - Business-as-usual scenario (BAS) as the baseline
 - Mitigation Scenario (2018- 2070)

Current Accounts: Final Energy Demand,2017

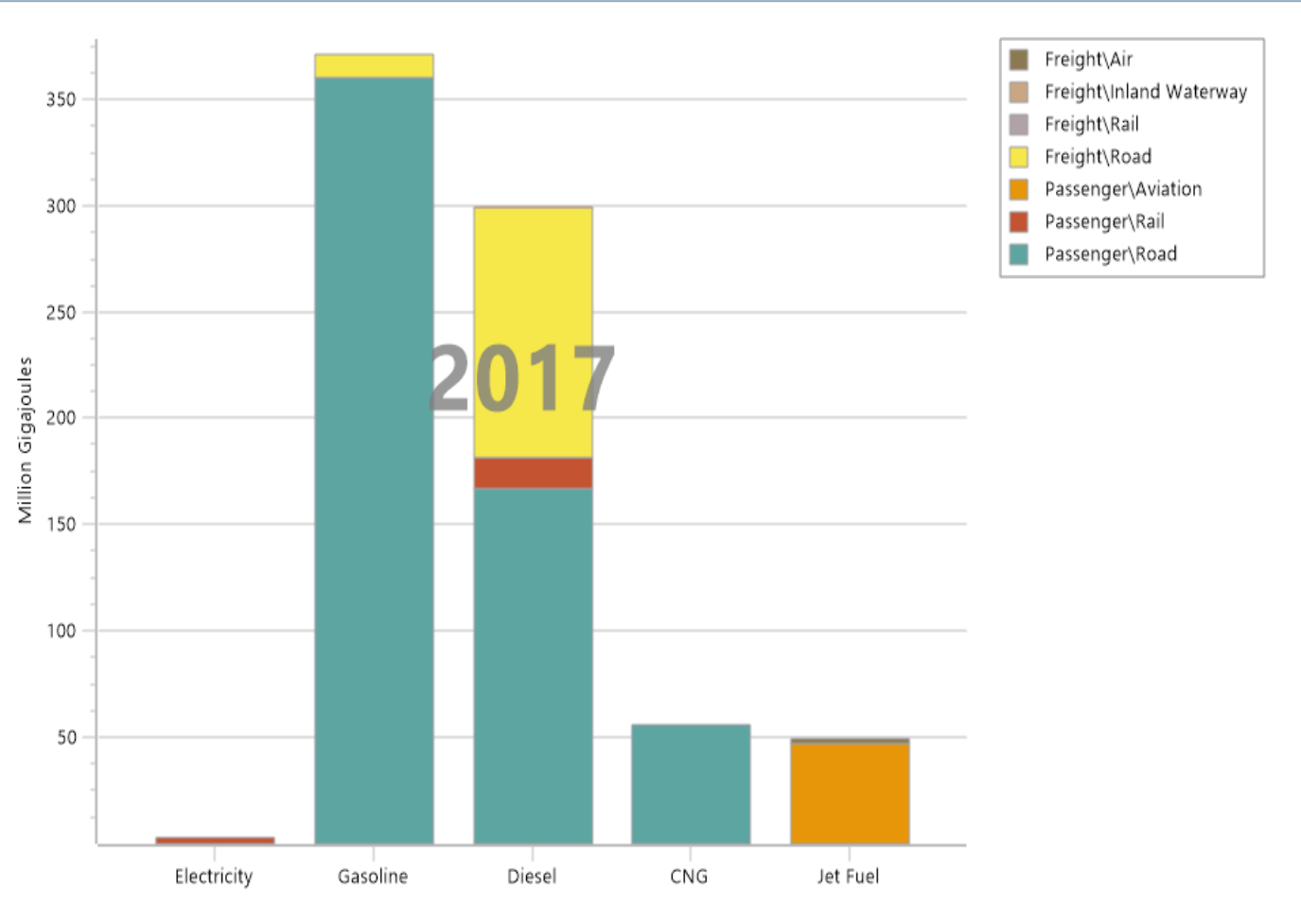
| Residential Sector



Final Energy Demand ,2017							
Branch: Demand: Residential Units: Million Gigajoules							
Branch	Electricity	Natural Gas	Kerosene	LPG	Solar	Biomass	Total
Urban Households	137.4	90.0	0.0	44.3	0.0	0.0	271.7
Rural Households	93.4	6.4	0.1	209.5	0.0	0.0	309.4
Total	230.9	96.4	0.1	253.8	0.0	0.0	581.1



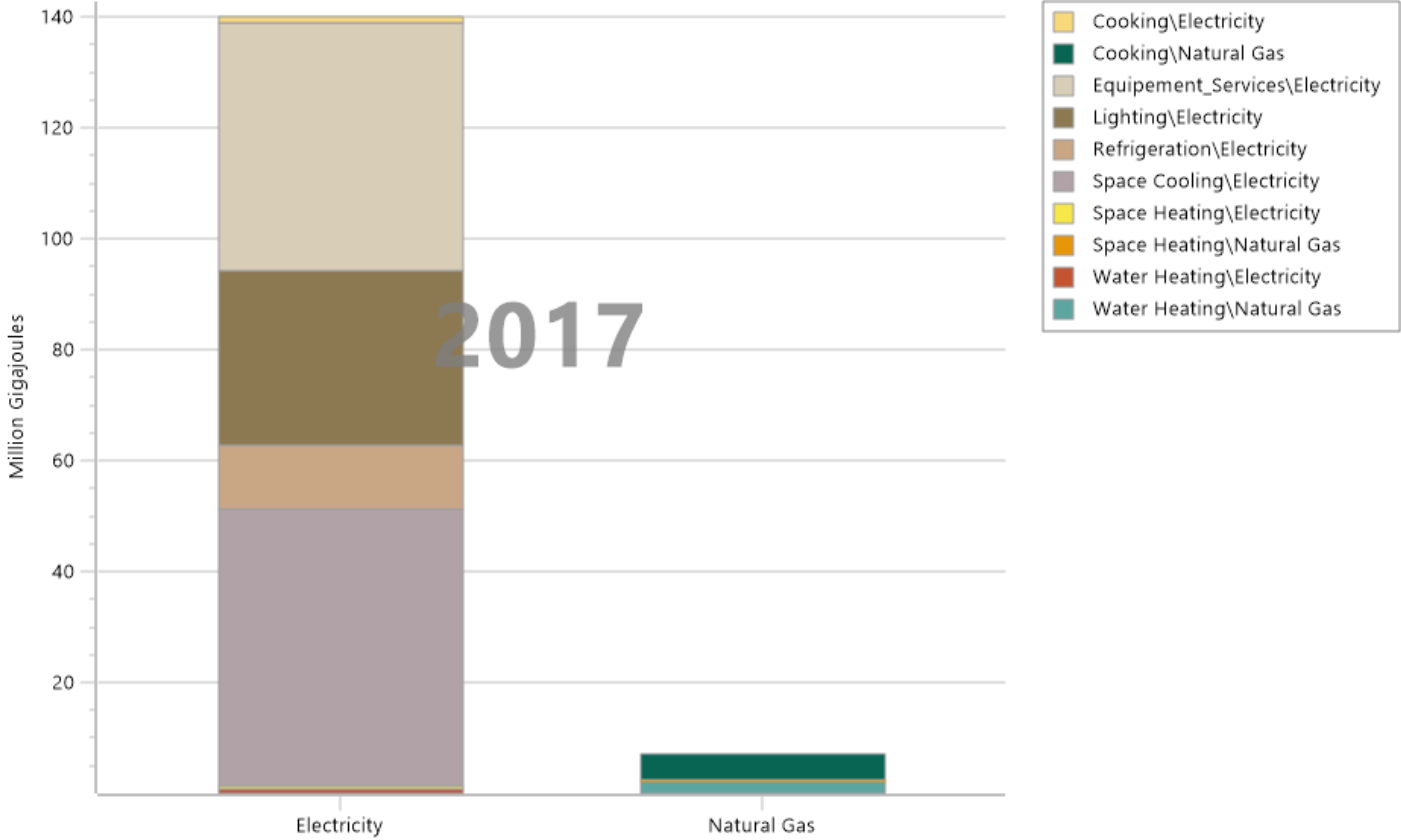
Final Energy Demand ,2017 Branch: Demand, Industry Units: Million Gigajoules							
Branch	Electricity	Natural Gas	Diesel	Residual Fuel Oil	Coal Bituminous	RDF	Total
Iron and Steel	5.5	7.9	-	0.0	0.2	-	13.7
Cement and Other Non_mettallic Minerals	26.1	33.1	4.5	10.0	126.7	10.8	211.1
Fertilizers	14.9	128.7	-	0.8	-	-	144.5
Petrochemicals and Basic Chemicals	0.5	20.4	-	3.9	-	-	24.8
Refining and Gas Processing	17.2	15.6	-	41.8	-	-	74.6
Food Processing and Beverages	14.1	20.3	0.9	2.0	-	-	37.3
Textiles and Apparel	0.1	0.1	-	0.0	-	-	0.2
Paper and Printing	4.8	5.6	-	1.5	-	-	11.9
Mining and Quarrying	4.1	-	0.0	-	-	-	4.2
Other industries and Manufacturing	29.3	76.4	-	56.9	-	-	162.5
Total	116.6	308.1	5.3	116.9	126.9	10.8	684.7



Final Energy Demand , Transport Sector ,2017						Units: Million Gigajoules
Branch	Electricity	Gasoline	Diesel	CNG	Jet Fuel	Total
Passenger	2.7	359.7	181.8	55.8	47.5	647.4
Road	-	359.7	167.3	55.8	-	582.7
Cars	-	321.2	11.4	22.1	-	354.7
Buses	-	-	37.8	1.1	-	39.0
Taxi	-	20.4	-	23.5	-	43.8
Motorcycles_Scooters	-	13.2	-	-	-	13.2
Microbuses	-	-	118.1	9.0	-	127.1
Tuk Tuks	-	4.9	-	-	-	4.9
Rail	2.7	-	14.5	-	-	17.2
Electricity _ Cairo Metro	2.7	-	-	-	-	2.7
Diesel	-	-	14.5	-	-	14.5
Aviation	-	-	-	-	47.5	47.5
Jet Fuel	-	-	-	-	47.5	47.5
Freight	-	11.1	117.6	-	1.6	130.3
Road	-	11.1	117.3	-	-	128.4
Light Trucks	-	11.1	14.9	-	-	26.0
Medium Trucks	-	-	22.4	-	-	22.4
Heavy Trucks	-	-	80.0	-	-	80.0
Rail	-	-	0.1	-	-	0.1
Diesel	-	-	0.1	-	-	0.1
Inland Waterway	-	-	0.1	-	-	0.1
Diesel	-	-	0.1	-	-	0.1
Air	-	-	-	-	1.6	1.6
Jet Fuel	-	-	-	-	1.6	1.6
Total	2.7	370.8	299.4	55.8	49.1	777.7

Current Accounts: Final Energy Demand,2017

| Commercial & Public Sector



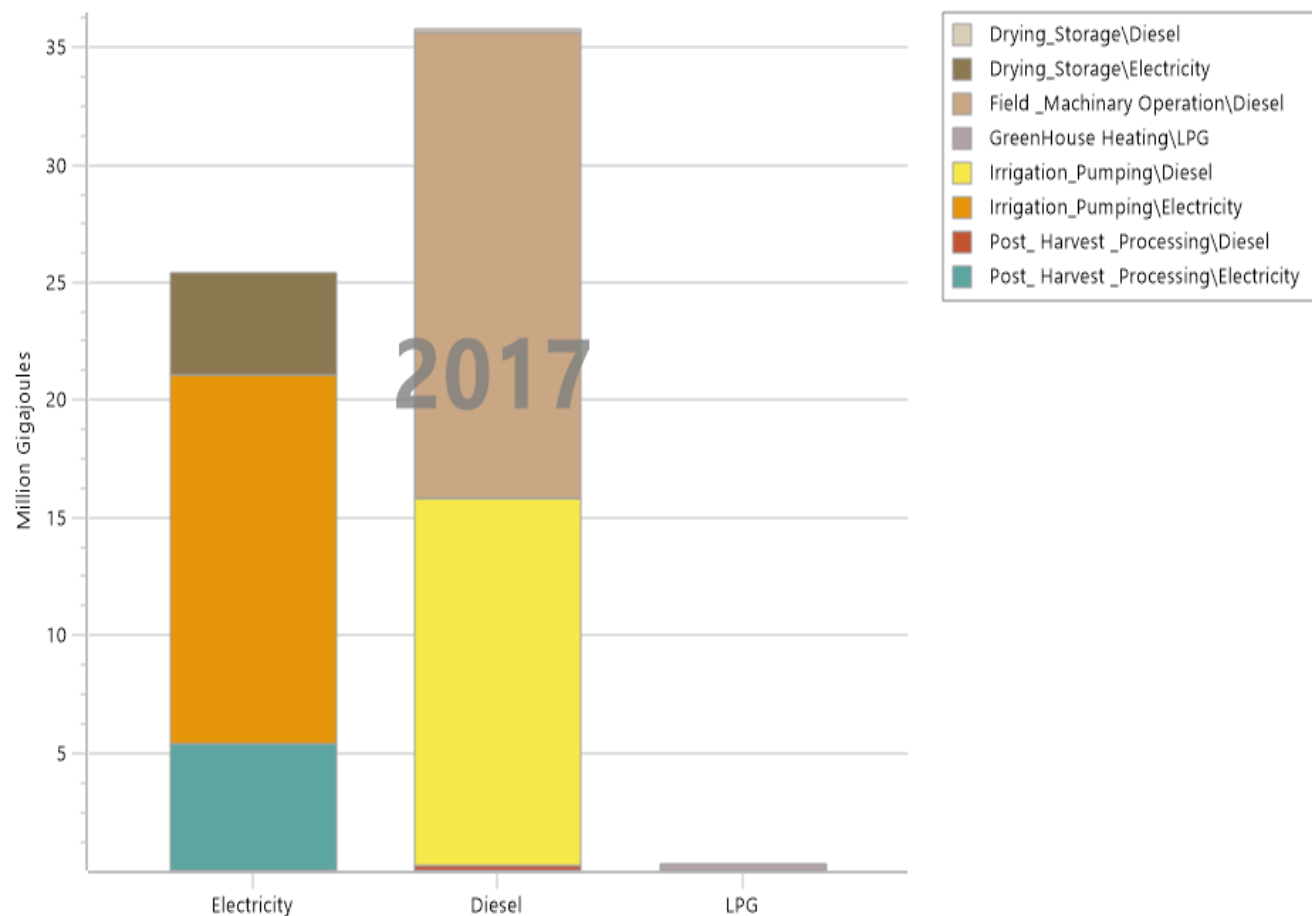
Final Energy Demand, Commercial _ Public Services, 2017

Units: Million Gigajoules

Branch	Electricity	Natural Gas	Total
Space Cooling	49.9	-	49.9
Electricity	49.9	-	49.9
Lighting	31.5	-	31.5
Electricity	31.5	-	31.5
Water Heating	0.8	2.3	3.0
Electricity	0.8	-	0.8
Natural Gas	-	2.3	2.3
Refrigeration	11.5	-	11.5
Electricity	11.5	-	11.5
Cooking	1.1	4.5	5.7
Electricity	1.1	-	1.1
Natural Gas	-	4.5	4.5
Space Heating	0.6	0.2	0.8
Electricity	0.6	-	0.6
Natural Gas	-	0.2	0.2
Equipement_Services	44.6	-	44.6
Electricity	44.6	-	44.6
Total	139.9	7.0	146.9

Current Accounts: Final Energy Demand,2017

| Agriculture Sector

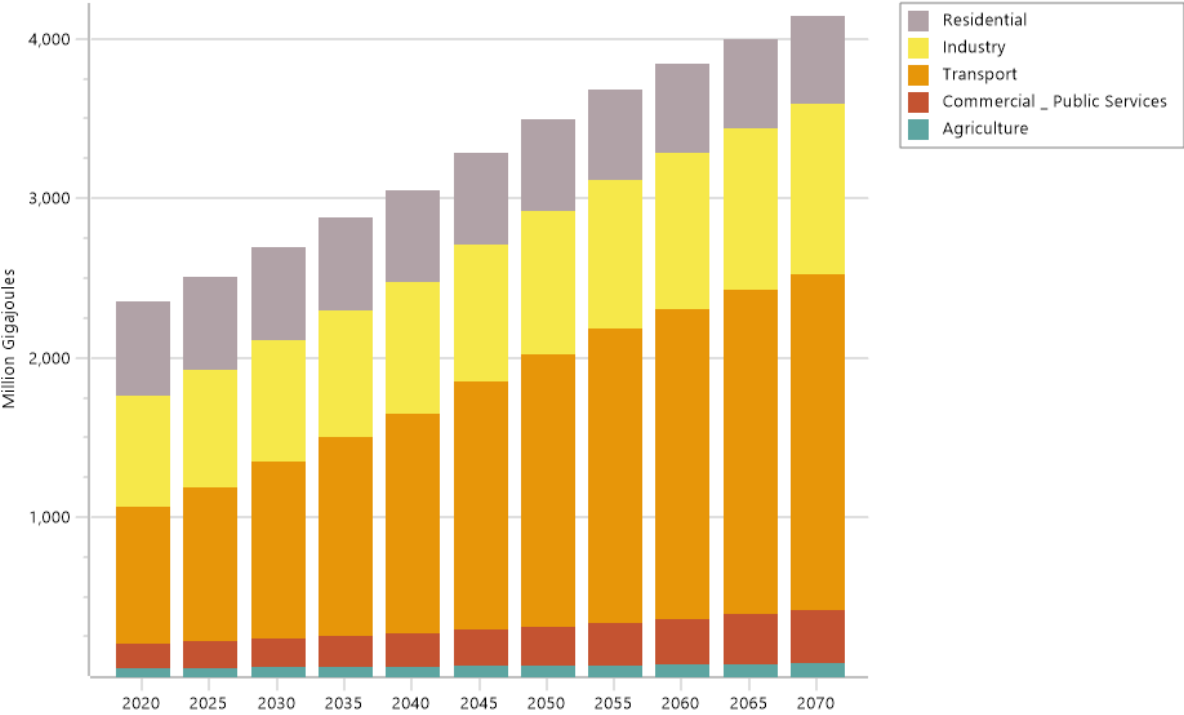


Final Energy Demand, Agriculture Sector 2017, Units: Million Gigajoules

Branch	Electricity	Diesel	LPG	Total
Irrigation_Pumping	15.7	15.7	-	31.3
Diesel	-	15.7	-	15.7
Electricity	15.7	-	-	15.7
Field_Machinery Operation	-	19.8	-	19.8
Diesel	-	19.8	-	19.8
Post_Harvest_Processing	5.4	0.2	-	5.6
Electricity	5.4	-	-	5.4
Diesel	-	0.2	-	0.2
Drying_Storage	4.4	0.1	-	4.5
Electricity	4.4	-	-	4.4
Diesel	-	0.1	-	0.1
GreenHouse Heating	-	-	0.3	0.3
LPG	-	-	0.3	0.3
Total	25.4	35.8	0.3	61.5

Final Energy Demand , Scenario: Baseline, All Fuels, Units: Million Gigajoules

Branch	2020	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070 Total	
Residential	581.5	579.5	577.3	574.6	571.7	568.6	565.4	560.3	554.9	549.4	543.6	6,226.7
Industry	702.0	731.5	762.2	792.6	825.0	859.6	896.6	934.9	976.2	1,020.9	1,069.2	9,570.8
Transport	859.7	971.9	1,118.7	1,253.7	1,377.6	1,563.2	1,713.3	1,846.7	1,947.1	2,035.9	2,108.6	16,796.3
Commercial _ Public Services	154.0	166.6	180.1	194.9	210.8	228.1	246.8	266.7	288.2	311.5	336.7	2,584.4
Agriculture	55.9	58.4	61.0	63.9	66.8	69.9	73.1	76.8	80.5	84.5	88.5	779.4
Total	2,353.2	2,507.8	2,699.3	2,879.7	3,052.0	3,289.5	3,495.2	3,685.3	3,847.0	4,002.1	4,146.7	35,957.6



Final Energy Demand by Fuel: (Million GJ)

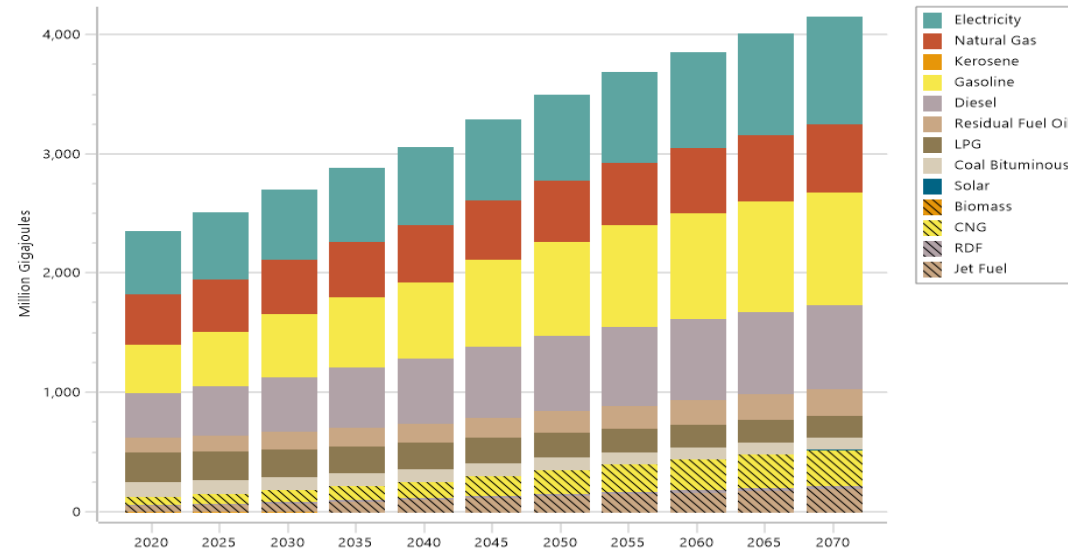
Scenario: Baseline



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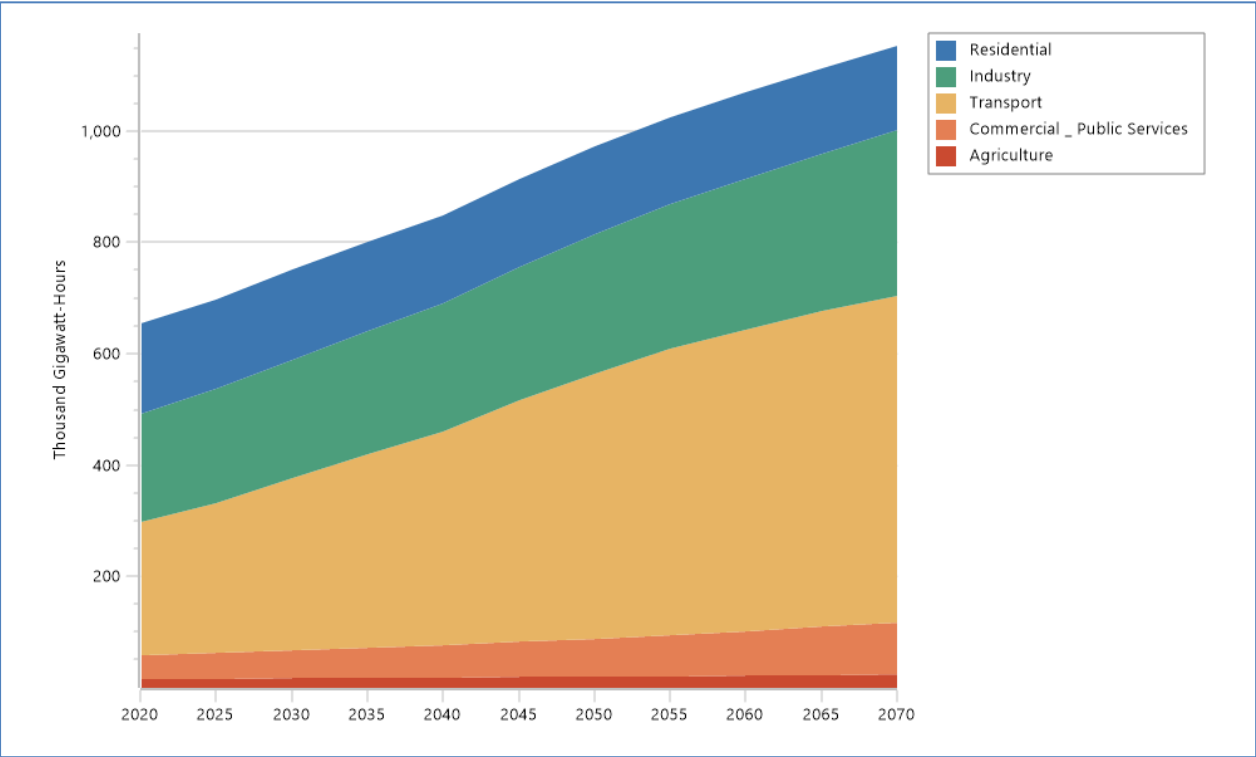
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Lab.Spatial Data Science



Final Energy Demand Scenario , Baseline Units: Million Gigajoules

Fuel	2020	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	Total
Electricity	523.6	551.5	581.2	610.2	641.5	676.0	713.2	752.3	794.7	841.9	893.5	7,579.5
Natural Gas	421.8	438.2	454.8	468.8	483.1	497.8	512.7	526.7	541.0	555.6	570.4	5,470.9
Kerosene	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	0.1
Gasoline	408.4	460.2	527.7	586.7	638.7	724.3	791.0	847.9	888.0	920.6	944.9	7,738.5
Diesel	371.1	409.6	459.8	504.0	543.6	593.4	629.9	660.6	679.9	692.9	700.3	6,245.2
Residual Fuel Oil	122.9	133.1	143.6	152.0	160.9	170.4	180.6	191.7	203.7	216.6	230.6	1,906.0
LPG	250.4	240.5	230.9	224.5	218.2	212.0	205.8	198.7	191.7	184.8	178.1	2,335.7
Coal Bituminous	124.1	119.3	114.6	112.5	110.3	108.2	106.0	105.4	104.8	104.1	103.4	1,212.6
Solar	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.5
Biomass	-0.0	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
CNG	63.7	77.2	94.2	113.7	133.6	165.8	196.6	225.3	251.2	277.6	302.5	1,901.2
RDF	11.5	12.8	14.0	15.8	17.5	19.3	21.0	22.2	23.3	24.4	25.4	207.1
Jet Fuel	55.6	65.3	78.4	91.5	104.5	122.4	138.3	154.5	168.7	183.5	197.5	1,360.3
Total	2,353.2	2,507.8	2,699.3	2,879.7	3,052.0	3,289.5	3,495.2	3,685.3	3,847.0	4,002.1	4,146.7	35,957.6

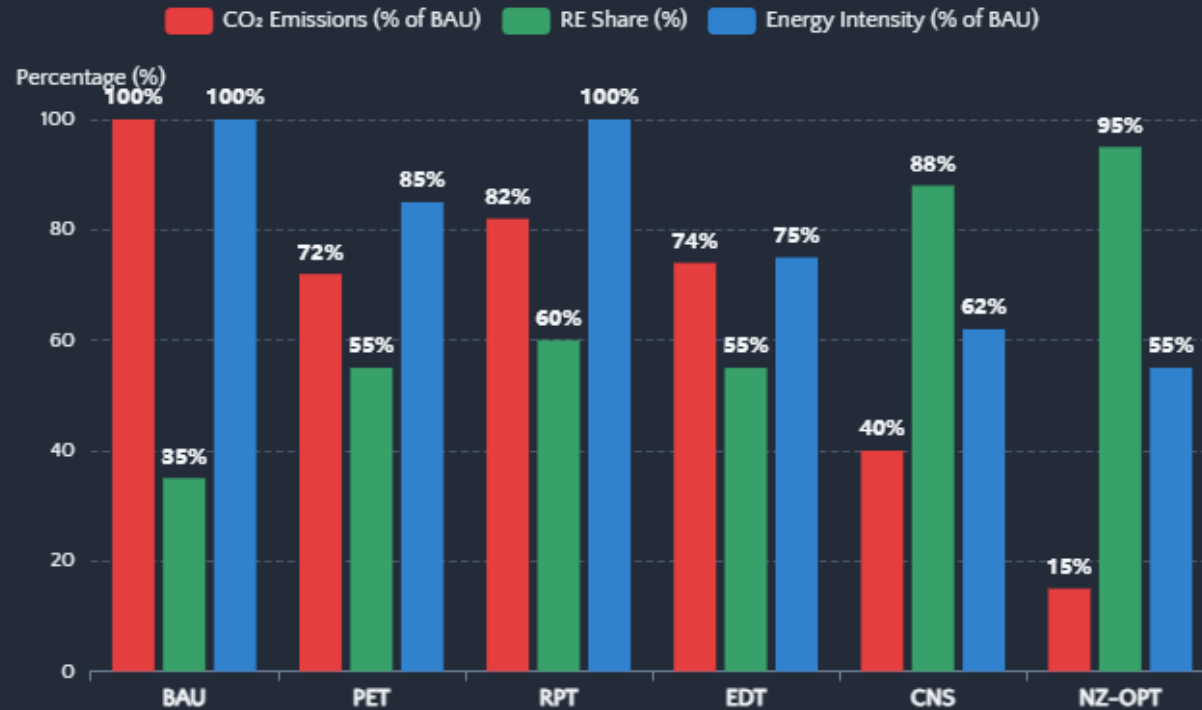
Final Energy Demand Scenario: Baseline, All Fuels



Final Energy Demand Scenario: Baseline, All Fuels , Units: Thousand Gigawatt-Hours												
Branch	2020	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	Total
Residential	161.6	161.0	160.4	159.7	158.9	158.0	157.1	155.7	154.2	152.7	151.1	1,730.3
Industry	195.0	203.2	211.7	220.2	229.2	238.8	249.1	259.7	271.2	283.6	297.0	2,658.5
Transport	238.8	270.0	310.7	348.3	382.7	434.2	475.9	513.0	540.9	565.5	585.7	4,665.6
Commercial _ Public Services	42.8	46.3	50.0	54.1	58.6	63.4	68.6	74.1	80.1	86.5	93.5	717.9
Agriculture	15.5	16.2	16.9	17.7	18.6	19.4	20.3	21.3	22.4	23.5	24.6	216.5
Total	653.7	696.7	749.9	800.0	847.8	913.8	971.0	1,023.8	1,068.7	1,111.7	1,151.9	9,988.9

Scenario Comparison: Key Expected Findings

Comparative Performance Indicators by Scenario (2070)



Demand-side efficiency has the greatest impact in residential & industrial sectors.



GHG trajectory
by scenario



Energy mix
evolution



Investment
requirements



Policy
recommendations

Key Insights

Neither renewables alone (RPT) nor efficiency alone (EDT) achieves **Carbon neutrality**; both must be combined for deep decarbonization.

EDT/RPT: Renewable-Led Power Transition Efficiency Slows Demand growth and reduce emissions, yet residual emissions remain from fossil fuel use.

CNS scenario shows ~60–75% reduction in CO₂ emissions vs BAU by 2070 through coordinated deployment.

Only the **CNS / Carbon-Neutrality Strategy + NZ-OPT/ NEMO Net-Zero** – Least-Cost Pathway, combined pathway enables deep decarbonization and a robust route to carbon neutrality.

Policy-aligned (PET) scenario underperforms on emissions without stronger electrification mandates.



Critical Finding

The **NZ-OPT benchmark** reveals that the least-cost pathway requires a **90%+ RE share and significant electrification of end-uses**.

Critical Findings for Egypt's Energy Transition



Combined Approach Essential

Clean electricity and efficiency each reduce emissions, but neither is sufficient alone. Achieving carbon neutrality requires coordinated deployment of both supply-side and demand-side measures.



Sector-Specific Efficiency Impact

Demand-side efficiency has the greatest impact in residential & industrial sectors. Buildings' energy intensity reduction (30–40% by 2050) and industrial efficiency improvements are key levers.



Rapid Demand Growth Challenge

Rapid energy-demand growth requires integrating supply-side and demand-side measures. With demand projected to triple by 2050, efficiency improvements are critical to moderate growth.



Policy Gaps in Current Plans

Policy-aligned (PET) scenario underperforms on emissions without stronger electrification mandates. Current official targets need enhancement to achieve deep decarbonization.



Core Policy Implication

Achieving carbon neutrality in Egypt depends on combining renewable expansion with strong efficiency policies. Carbon neutrality requires coordinated action across both energy supply and demand - a single-strategy approach is insufficient for deep decarbonization.

Conclusion & Research Contributions

✓ Key Conclusions

- Egypt's long-term energy transition cannot rely on a single strategy. Renewable electricity and energy efficiency each play essential but incomplete roles in achieving carbon neutrality.
- An integrated, LEAP-based scenario framework provides a realistic pathway toward carbon neutrality by 2070, demonstrating that coordinated transformation is achievable.
- Carbon neutrality in Egypt is achievable only through a coordinated, long-term transformation of Both energy supply and demand are supported by strategic investment sequencing.

“ This study delivers the most comprehensive long-range energy system model for Egypt, demonstrating that carbon neutrality by 2070 is achievable but requires coordinated deployment of renewables, demand-side efficiency, and electrification.

★ Key Research Contributions

1 First Integrated Model

First integrated LEAP+NEMO model calibrated to Egypt's 2017 energy system, providing a robust analytical foundation.

2 Structured Framework

Structured 6-scenario framework from BAU to the least-cost net-zero benchmark, enabling systematic policy assessment.

3 Long-Range Analysis

Long-range analysis extending to 2070 — beyond all existing Egyptian studies, providing unprecedented temporal scope.

4 Transferable Methodology

Transferable modeling methodology for other MENA & developing economies facing similar energy-climate challenges.

Thank You

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